

Algorithm block deployed to ACCURATE marketplace

Aliaksandr Nekrashevich^a, Valeria Borodin^a, Alexandre Dolgui^a, Simon Thevenin^a

^a*Department of Automation, Production and Computer Sciences, IMT Atlantique Bretagne - Pays de la Loire, LS2N, Campus de Nantes, 4 rue Alfred Kastler - La Chantrerie, Nantes, Cedex 3, CS 20722, 44307, France*

Abstract

In this document, we describe the block deployed to the ACCURATE marketplace through Pontus-X.

1. Introduction

This algorithm asset focuses on simultaneous dynamic pricing and scheduling within a Manufacturing as a Service (MaaS) context. Active revenue management implies the application of dynamic pricing, and standard techniques become less applicable in an industrial context due to the combinatorial nature of scheduling and the continuous nature of the dynamic pricing component, even from the viewpoint of an individual MaaS serving customer demand. In this work, we address this gap by jointly considering dynamic pricing and scheduling, relying on genetic heuristics, sampling, and integrated optimization.

2. Input Format

Input is provided in JSON format, through any format supported by Pontus-X, for example, as a reference to a raw file in a GitHub repository. Explicit example: <https://raw.githubusercontent.com/imt-nekrald/raw-data/refs/heads/main/testdata.json>.

The input file specifies three blocks: **request** block, **parameters** block, and **configuration** block. In the **request** block, the type of requested operation is described. Three types of requests are supported: lift, price, and allocation.

2.1. Parameters block

The **parameters** block specifies the algorithm parameters, including configuration for the genetic algorithm and the number of trajectories during evaluation. The block is available through the key "parameters". The following information is available inside the block:

Block Description	Key
Genetic Algorithm Configuration	genetic-config
General-Purpose Parameters	general-parameters

The general-purpose parameters have the following keys and meanings:

Parameter	Key
Number of trajectories at final evaluation	evaluation-sample-count
Number of trajectories for rollout estimation	rollout-sample-count
Time limit for perfect info estimator	estimator-time-limit
Number of trajectories for nested estimation (ignored at present version)	nested-sample-count

The genetic algorithm configuration contains the following parameters:

Parameter	Key
Number of iterations in genetic algorithm	n-genetic-iterations
Number of places for mutation	n-mutation-places
Number of trajectories at evaluation	n-evaluation-trajectories
Number of static price tables after selection	selection-size
Number of trajectories for fast estimation at intermediate selection	n-fast-trajectories
Recombination probability for each feature	recombination-proba

2.2. Configuration block

The **configuration** block describes the parameters of the service provider: number of available machines, supported types of jobs, allowed price ranges, etc. The complete specification follows:

Key	Type	Description
n-machines	integer	Number of available machines
n-order-types	integer	Number of possible order types
n-periods	integer	Total number of time periods
order-types	list[Order Description]	List with order description, each order description is a dictionary
period-durations	list[double]	List with durations of periods
period-order-probability	list[list[double]]	Table listing probabilities to receive each order type at each period
price-ranges	list[list[list[double]]]	Price ranges, i.e. possible prices, indexed by [period][order-type]

Fields in the order type description:

Key	Type	Description
idx-type	integer	Index of order type
compatible-machines	list[integer]	Indices of compatible machines
count	integer	Number of jobs in the order
release-term	double	Term after order before availability
duration-per-machine	list[double]	Processing times per machine
delivery-per-machine	list[double]	Delivery times per machine
due-term	double	Term before due after order
tardiness-penalty	double	Tardiness fee per period of delay
decline-penalty	double	Penalty for late decline after accepting
linear-sensitivity	double	Sensitivity in the linear willingness-to-pay model

2.3. Request block

The block is available through the key "request". The first field to look in the corresponding sub-block is the type of request, described by the key "request-type".

Request Type	Description
lift	Estimates potential lift in described configuration. Allows to decide on relevance for particular configuration
price	Suggests price at the current period according to current state
allocation	Suggests allocation of order jobs to machines

For pricing and allocation requests, additional information is supplied.

Key	Type	Description
order-description	Order Description	Description of the order
period-id	integer	Period of the order
production-state	Production State (optional)	Description of production state

The format for order type description has been described above. Production state is optional, without this parameter, model assumes all production capacity is available. The format of production state is the following:

Key	Type	Description
state-time	double	Production state time point
allocated-revenue	double	Currently accumulated revenue among tasks already allocated
state-period	integer	Index of state time period
flexible-tasks	list[SchedulingTask]	Tasks currently accepted, but not yet allocated
allocated-tasks	list[SchedulingTask]	Tasks accepted and allocated
declined-tasks	list[SchedulingTask]	Tasks accepted but later declined
machine-availability	dict[int, double]	Dictionary representing machine availability times
machine-keys	list[int]	List representing machine indices
task-allocation	list[list[SchedulingInterval]]	Table describing task allocation
request-history	list[list[SchedulingTask]]	History of tasks requested per period
price-history	list[double]	History of prices per revealed periods
period-history	list[int]	Prefix of indices for revealed periods

Specification for scheduling task:

Key	Type	Description
uuid	string	Task identifier
price	double	Price per task
delay-cost	double	Tardiness fee
decline-cost	double	Decline penalty
compatible-machines	list[integer]	Indices of compatible machines
duration-per-machine	list[double]	Processing duration per machine
delivery-per-machine	list[double]	Delivery terms per machine
release-time	double	Time when the task is available
release-term	double	Term from order to release
due-time	double	Due time, when the task should be ready

Specification for scheduling interval:

Key	Type	Description
task	Scheduling Task	Specification of the scheduling task
start-time	double	Time of processing start
completion-time	double	Time of processing completion
delay-penalty	double	Tardiness cost paid
machine-id	int	Identifier of machine processing

For pricing requests, the following additional information is available:

Key	Type	Description
price-policy	Price Policy	Name of pricing policy, static or roll-out
static-table	list[list[double]]	Static pricing table (e.g. computed at the lift request)

For allocation requests, the following additional information is available:

Key	Type	Description
accumulated-time	double	Currently considered time point at production line

3. Output Format

The main output is provided in the JSON format and depending on the request, has the following format.

Allocation. For allocation response:

Key	Type	Description
production-state	Production State	Production state after allocation

The format of production state is the same as described in the input section.

Pricing. For price response:

Key	Type	Description
price	double	Price to suggest to the client
production-state	Production State	Production state after pricing

Lift estimaton. For lift request, the response has the following form:

Key	Type	Description
static-table	list[list[double]]	Static table to base static or rollout pricing
static-result	Lift Estimation Result	Structure containing description of the estimated lifts at static policy
rollout-result	Lift Estimation Result	Structure containing description of the estimated lifts at rollout policy

The lift estimation result has the following structure:

Key	Type	Description
profit-average	double	Estimated expected profit value
profit-lower-bound	double	Lower end of 95% profit confidence interval
profit-upper-bound	double	Upper end of 95% profit confidence interval

The Pontus-X evaluation block also provides log files at response. Reading the logs may prove beneficial for debugging.

4. Input Example

```
{
  "request ":
  {
    "request-type": "lift "
  },
  "parameters ":
  {
    "genetic-config ":
    {
      "n-evaluation-trajectories" : 10,
      "n-fast-trajectories" : 5,
      "n-genetic-iterations" : 3,
      "n-mutation-places" : 10,
      "recombination-proba" : 0.5,
      "selection-size" : 10
    },
    "general-parameters ":
    {
      "evaluation-sample-count": 10,
      "rollout-sample-count": 5,
      "estimator-time-limit": 500
    }
  },
  "configuration ":
  {
    "n-machines" : 1,
    "n-order-types" : 1,
    "n-periods" : 10,
    "order-types" :
    [
      {
        "compatible-machines" : [ 0 ],
        "count" : 10,
        "decline-penalty" : 50.0,
        "delivery-per-machine" : [ 3.0 ],
        "due-term" : 24.0,
        "duration-per-machine" : [ 0.25 ],
        "idx-type" : 0,
        "linear-sensitivity" : 0.04,
        "release-term" : 12.0,
        "tardiness-penalty" : 0.25
      }
    ]
  }
}
```

